

PROJECT CASE STUDY: Delivery Performance & DPMO Tracker

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Platform: Amazon Mercury (Internal Dashboard & Reporting Engine)

Project Overview

This project was developed to address a critical visibility gap in Last-Mile delivery operations. This custom-built Mercury dashboard provides real-time, high-definition tracking of undelivered packages, calculates the Defects Per Million Opportunities (DPMO) dynamically, and categorizes delivery failures to drive operational improvements.

The Problem

Lack of Real-Time KPI Tracking: Leadership and Shift Assistants lacked a centralized, real-time view of daily delivery health and DPMO metrics.

No Centralized Root-Cause Analysis: When packages were missed, digging into the "why" (e.g., vehicle breakdowns, customer-unreachable, bad addresses) required manually pulling and cross-referencing multiple data sources, delaying course corrections during active shifts.

The Solution

I built a real-time tracking dashboard utilizing Amazon's internal Mercury platform to ingest, process, and visualize delivery success metrics.

Dynamic DPMO Calculation: Automatically calculates delivery quality metrics. For example, if 1,000 packages are scheduled and 990 are successfully delivered, the dashboard dynamically calculates and displays a DPMO of 10,000.

Granular Failure Breakdown: Features a drill-down data table listing every single missed package, mapping it to the assigned route, driver, and specific standard failure code.

Visual Pareto Charting: Categorizes delivery failures (e.g., Access Issue, Business Closed, Package Damaged) to visually show operations teams exactly where to focus their recovery efforts.

System Architecture & Data Flow

[Raw Delivery Data Stream] —► Ingests live logistics data.

[Mercury Engine] —► Processes custom query logic and filtering.

[Aggregation Layer] —► Calculates DPMO: $(\text{Defects} / \text{Opportunities}) * 1,000,000$.

[Dashboard UI] —► Displays KPI cards, Pareto breakdowns, and the active defect log.

Business Impact & KPIs

Improved Shift Response: Reduced the time to identify and action failing routes from hours to minutes, allowing for mid-shift package rescues.

Targeted Feedback: Provided Process Assistants and Managers with clear, driver-level data to address recurring delivery issues.

Precision Quality Tracking: Transitioned the site from relying on basic percentage metrics to utilizing industry-standard DPMO, enabling highly precise quality control.